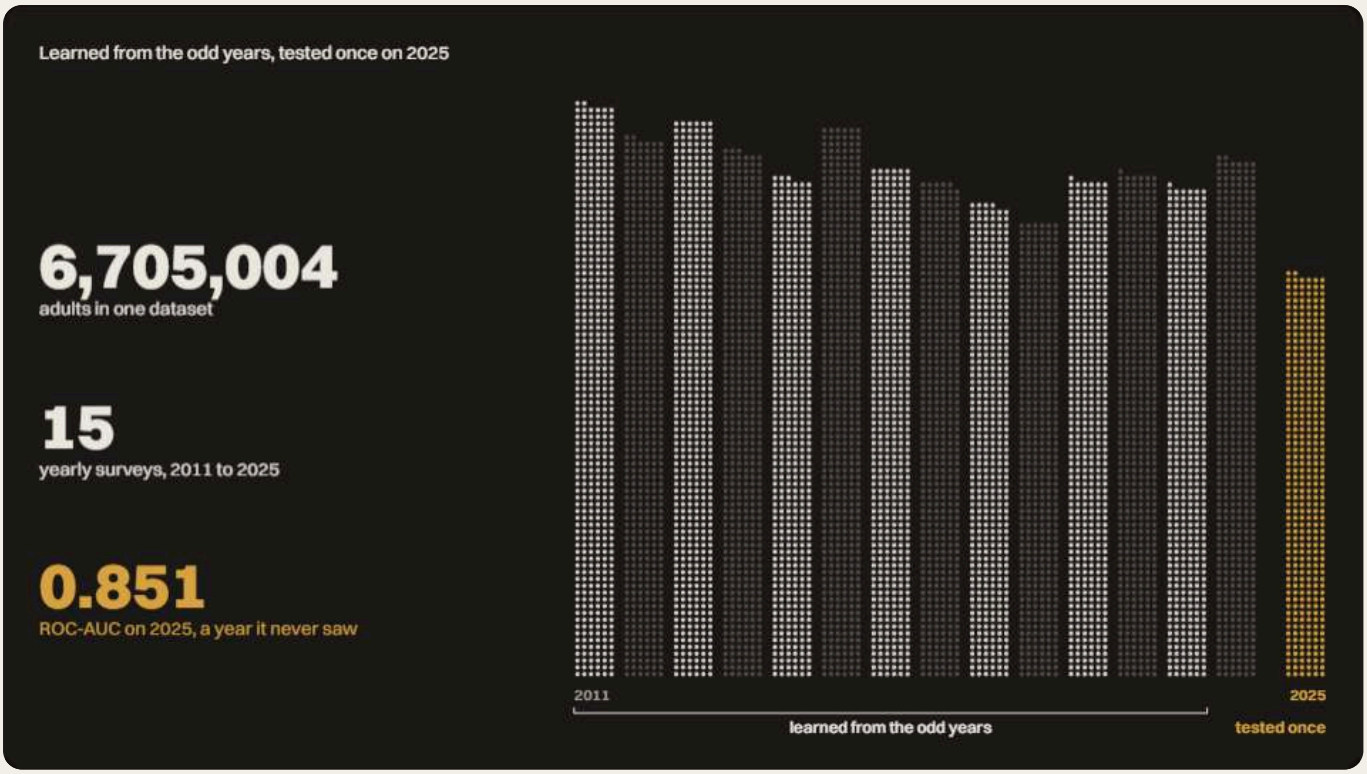
# Heart Risk Model

15 years of US surveys, tested on a year it never saw




THE PROBLEM

Can 15 years of US health surveys estimate a person's heart-disease risk honestly?

WHAT I BUILT

One dataset from 15 yearly CDC files (6.7 million answers), a model trained on past years and a public what-if app.

THE RESULT

ROC-AUC 0.851 on 2025, a year the model never saw.

**0.851**
ROC-AUC on 2025: 352,145 people the model never saw

**0.376**
PR-AUC on 2025, 4 times the no-skill level of 0.090

**6.3%**
heart disease, age-adjusted, in 2025, down from 7.1% in 2011

**6.7M**
adults in one dataset, from 15 yearly files

TOOLS

Python · pandas · scikit-learn · Streamlit · pytest · GitHub Actions

LINKS

Full project page
mellithyy.github.io/work/heart

Live app
heart-disease-risk-mellithyy.streamlit.app

Code on GitHub
github.com/mellithyy/heart-disease-risk-model